MINIPROJECT ENHANCEMENT REPORT

Ticket Management System - Enhancement Report

Authentication, SLA Prioritization, Visual Analytics & Dashboard Notifications

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ABSTRACT

This report documents the proposed backend enhancements for the Ticket Management System. The enhancement scope focuses on strengthening account security, improving operational ticket prioritization through SLA-aware logic, introducing automated progress reporting through scheduled emails with generated PDF attachments, converting numerical ticket data into graphical visualizations, and notifying the Admin dashboard when new tickets are raised or ticket statuses change.

The enhancements span five major domains: (1) Email-Based Two-Factor Authentication (2FA), CAPTCHA Verification, and secure Forgot-Password Recovery for both Admin and User accounts; **(2) SLA-Based ticket prioritization that ensures tickets nearing breach are resolved before newer non-critical tickets;** (3) Automated progress emails every 10-minutes with a PDF report containing ticket IDs, statuses, progress updates, and timestamps; (4) graphical data visualization that converts numerical ticket data into charts for better understanding; and (5) Admin dashboard notifications for newly raised tickets and ticket status updates.

The implementation is designed to extend the current backend without breaking existing APIs, database behavior, or the established UI/UX. Required database changes are additive, email, PDF, visualization, and notification services are modularized, and background scheduling is introduced with safeguards to avoid duplicate or skipped notifications.

1. ENHANCEMENT OVERVIEW

1.1 Background

The ticket system already supports ticket creation, ticket status management, and user/admin workflows. The SLA-based prioritization enhancement focuses only on improving operational ticket handling so that work queues are ordered by deadline risk instead of simple newest-first behavior.

1.2 Enhancement Goals

- Prioritize tickets by SLA urgency so tickets close to breach are handled first.
- Prevent newer non-critical tickets from overriding older or more urgent SLA-critical tickets.
- Keep the existing ticket creation flow unchanged and apply prioritization only during listing, assignment, or resolution queue handling.
- Maintain backward compatibility by adding optional SLA fields only where useful.

1.3 Enhancement Summary Table

SLA-Based Prioritization: Deadline-aware sorting, SLA-critical ticket escalation, no change to ticket creation flow.

2. ENHANCEMENT: SLA-BASED TICKET PRIORITIZATION LOGIC

The ticket handling logic is enhanced so that SLA-critical tickets are resolved before newer tickets. This ensures operational fairness is based on deadline risk rather than simple creation order alone.

2.1 Prioritization Rule

- Each ticket is evaluated using its SLA deadline or expected closure time.
- Tickets close to SLA breach are assigned the highest resolution priority.
- Newer tickets must not override older tickets that are near SLA breach.
- Existing ticket creation flow remains unchanged; prioritization is applied when fetching or assigning work.

2.2 Sorting Logic

Near SLA breach: Resolved first regardless of ticket creation time.

Already breached: Escalated to the top and flagged for immediate handling.

Normal active ticket: Sorted by configured priority and creation time after SLA-critical tickets.

Closed/resolved ticket: Excluded from active resolution queue unless reporting requires it.

2.3 Compatibility Requirements

- No change is made to the ticket creation request payload unless additional SLA metadata is required.
- Database changes are additive only, such as SLA deadline, SLA status, or last prioritization timestamp fields.
- Existing APIs continue to return compatible responses, with optional SLA fields added where useful.
- Business logic is isolated in a ticket prioritization service for clean testing and future modification.

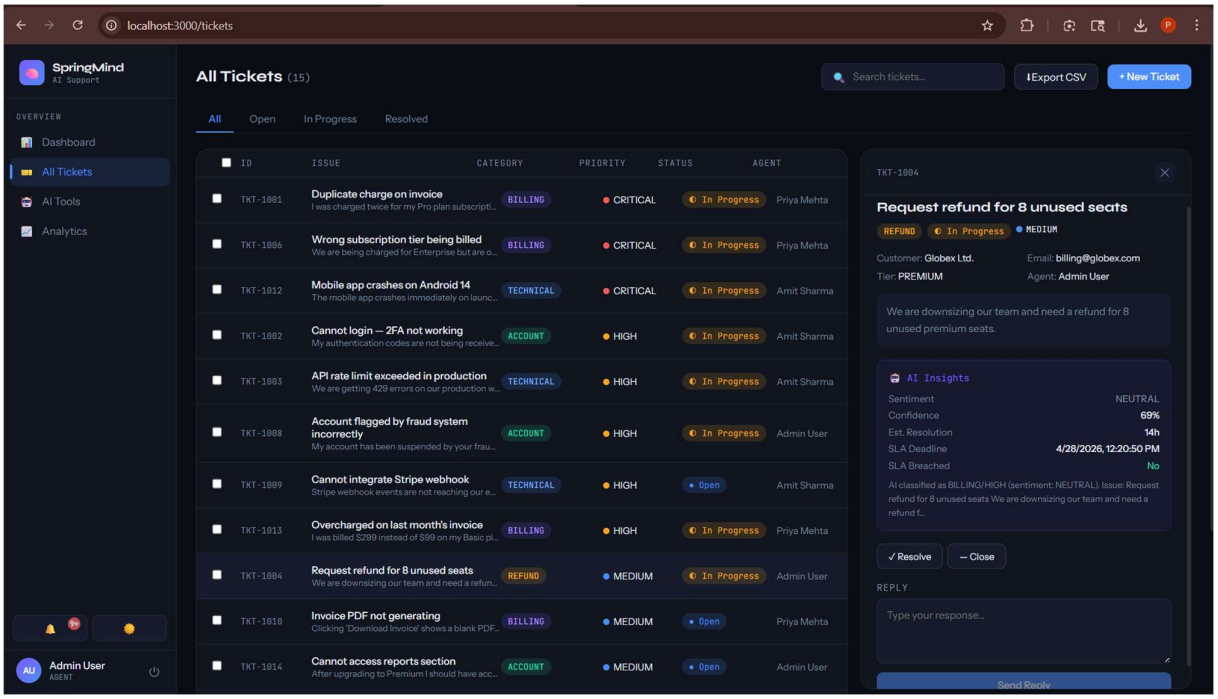

Figure 1: New Placeholder for SLA-prioritized ticket queue screenshot

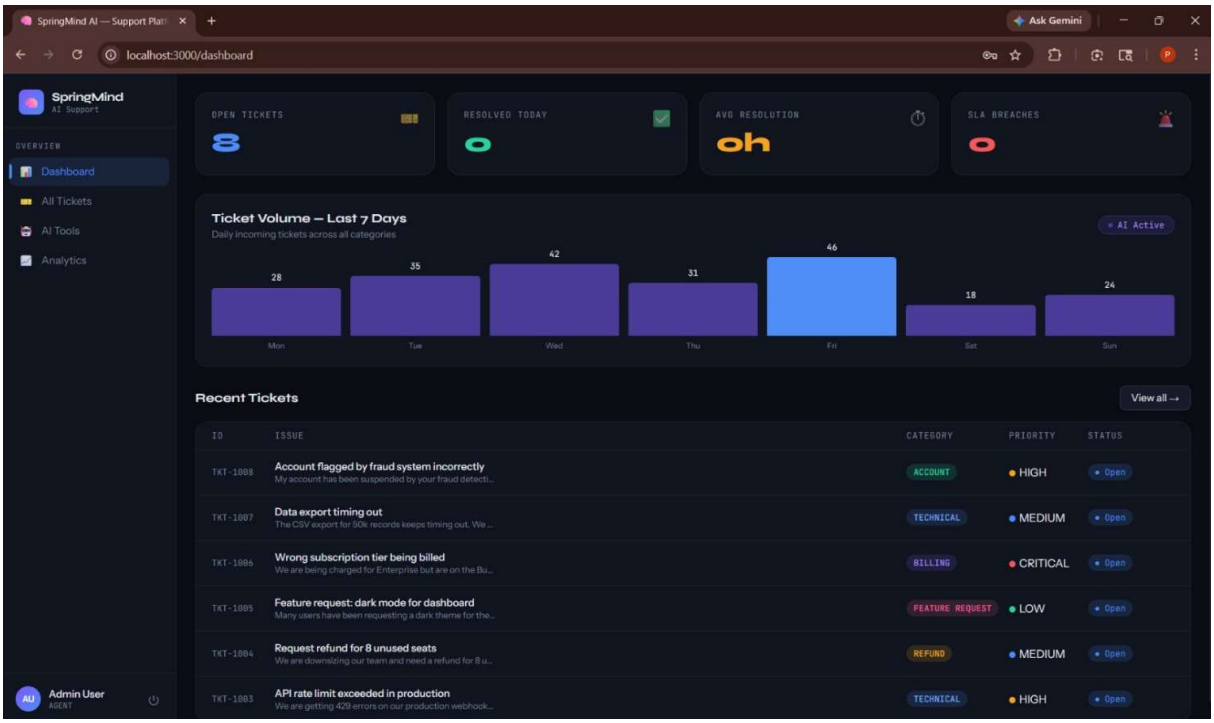

Figure 2: Old Placeholder for SLA-prioritized ticket queue screenshot

3. OUTCOME & RESULTS

3.1 Expected Features Delivered

- SLA Prioritization: Near-breach tickets are sorted above newer non-critical tickets.
- Resolution Queue Discipline: Agents see SLA-critical work first, reducing avoidable SLA breaches.
- Backward Compatibility: Existing APIs, UI flows, and ticket creation behavior continue to work after deployment.

3.2 Validation Checklist

- Tickets close to SLA breach remain ahead of newer tickets in resolution queues.
- Already breached tickets are escalated and clearly flagged.
- Closed or resolved tickets do not interfere with active SLA resolution ordering.
- Existing ticket creation behavior continues to work without request payload changes.

4. CONCLUSION

This enhancement strengthens the Ticket Management System by introducing SLA-aware ticket handling. The improved queue ordering ensures tickets nearing breach are resolved before newer non-critical tickets, helping agents respond according to business urgency and deadline risk.
The proposed design follows clean code practices by separating ticket prioritization into focused backend logic. Database updates remain additive and migration-driven, preserving backward compatibility while enabling reliable SLA-based queue behavior.

Learning Outcomes / Implementation Focus

- SLA-driven queue ordering that aligns system behavior with business urgency.
- Backward-compatible backend extension using modular services and additive migrations.
- Priority handling that prevents newer tickets from overriding near-breach tickets.